WORKED SOLUTION

STEP I 1995, Question 9

Mechanics • Projectile motion

> **STEP I 1995, Question 9**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/95-s1-q9

The key idea

Fix a point (x, y) and ask: is there an angle of projection whose trajectory passes through it? The trajectory equation, written in terms of $\tan\theta$ rather than θ, is a *quadratic* in $\tan\theta$. A point can be reached exactly when that quadratic has a real root, so the whole question comes down to a discriminant.

Solution

The model

We treat the particle as a point mass moving under gravity alone: no air resistance, g constant in size and direction, and nothing in the way of the path. The particle then moves in the vertical plane containing its initial velocity, and since only the horizontal *distance* from O matters, by symmetry we may take $x \geq 0$, with the particle projected towards increasing x at angle θ above the horizontal, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. (Straight up, $\theta = \frac{\pi}{2}$, is dealt with at the end.)

The trajectory

Write $u = \sqrt{2gh}$, so $u^2 = 2gh$. With constant acceleration g downwards, after time t

$$x = ut\cos\theta, \qquad y = ut\sin\theta - \tfrac{1}{2}gt^2.$$

Eliminating $t = x/(u\cos\theta)$ and using $\sec^2\theta = 1 + \tan^2\theta$,

$$y = x\tan\theta - \frac{gx^2}{2u^2}(1 + \tan^2\theta) = x\tan\theta - \frac{x^2}{4h}(1 + \tan^2\theta),$$

since $2u^2 = 4gh$. Writing $T = \tan\theta$ and multiplying through by $4h$, the trajectory passes through (x, y) if and only if

$$x^2T^2 - 4hx\,T + (x^2 + 4hy) = 0. \qquad (*)$$

As θ runs over $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $T = \tan\theta$ takes every real value exactly once, so choosing an angle is the same as choosing a real number T.

Ask yourself Why write the trajectory in terms of $\tan\theta$ instead of θ?

In terms of θ the equation mixes sin, cos and $\sec^2$, which is awkward. With $\sec^2\theta = 1 + \tan^2\theta$ everything is a polynomial in the one variable T, and "is there an angle?" becomes "does this quadratic have a real root?", a question we can answer at once.

Which points can be reached

Take $x > 0$. Then $(*)$ is a genuine quadratic in T, and it has a real root if and only if its discriminant is non-negative:

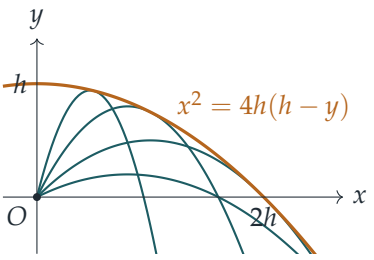
$$\begin{aligned}(4hx)^2 - 4x^2(x^2 + 4hy) \geq 0 &\iff 4x^2(4h^2 - x^2 - 4hy) \geq 0 \\ &\iff x^2 \leq 4h^2 - 4hy = 4h(h-y),\end{aligned}$$

where we divided by $4x^2 > 0$. So if (x, y) lies *above* the parabola, meaning $x^2 > 4h(h-y)$, or equivalently $y > h - \dfrac{x^2}{4h}$, then $(*)$ has no real root and no angle of projection sends the particle through (x, y).

It remains to deal with $x = 0$, the vertical line through O. Points on it with $y \neq 0$ are reached only by projecting vertically (any other angle gives $x > 0$ for all $t > 0$). Projected straight up with speed u, the particle rises to height $u^2/(2g) = h$ and no further, which is exactly the point where the parabola meets $x = 0$. So on this line too nothing above the parabola is reached.

Hence, whatever the angle of projection, the particle can never reach a point above the parabola $x^2 = 4h(h-y)$. ■

The bound is sharp: when $x^2 = 4h(h-y)$ the discriminant is zero and there is exactly one angle, $\tan\theta = 2h/x$, whose path just touches the parabola. The parabola is the *envelope* of all the trajectories, often called the parabola of safety.



Four trajectories (teal) with speed $\sqrt{2gh}$, each touching the boundary parabola (amber) once.

Simplifying assumptions

- The particle is a point mass (no size, no spin).
- Air resistance (and any other force except gravity) is neglected.
- The gravitational field is uniform: g is constant over the region of motion, which also means treating the Earth's surface as flat and ignoring its rotation.
- The path is unobstructed, for instance by the ground, so the particle can actually travel to the points in question.

Checks

- At $x = 0$ the parabola gives $y = h = u^2/(2g)$, the greatest height of a vertical throw. On $y = 0$ it gives $x = 2h = u^2/g$, the familiar maximum range on level ground (at 45°, where $\tan\theta = 2h/x = 1$, as predicted).
- Numerically, with $g = 9.8$ and $h = 1.7$, we stepped the motion for 399 launch angles from almost straight down to almost straight up; the greatest height above the parabola anywhere on any path was about -5×10^{-13}, i.e. zero up to rounding, and maximising y over angle at fixed x reproduced $h - x^2/(4h)$ every time.

Common slips

- Working with θ in the range $\left[0, \frac{\pi}{2}\right]$ only. Points below O need downward projections ($T < 0$), and the argument has to cover them.
- Dividing by x (or x^2) without treating $x = 0$ separately.
- Showing only that one particular trajectory lies under the parabola; the claim is about *every* angle, which is why we fix the point and vary the angle.

Beyond STEP

The parabola here is the *envelope* of the one-parameter family of trajectories: the curve touching every member. For a family $F(x, y, T) = 0$ the envelope is found by eliminating T between $F = 0$ and $\partial F/\partial T = 0$, which for a quadratic in T is precisely the vanishing of the discriminant. Envelopes reappear in caustics in optics, in the theory of first-order differential equations (singular solutions) and in the "safe region" for all projectiles of a given speed. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for projectile motion with and without air resistance.

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